

Use-Case Flow Specification

Lab 1 — Requirements Engineering & UML Use-Case Modelling
Problem Statement #53 — Freelance Content Creator Escrow Platform

Name	SRN
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UC-04 — Release Milestone Payment

Field	Value
Use Case ID	UC-04
Use Case Name	Release Milestone Payment
Primary Actor	Client Sponsor
Secondary Actors	Content Creator (notified), Payment Gateway (external system)
Related Requirements	FR-001, FR-002, NFR-002
Includes	UC-08 Authorize Escrow Transfer
Extended by	UC-05 Raise Dispute
Trigger	The Client Sponsor selects "Approve & Release" on a milestone whose draft is in <code>SUBMITTED</code> state.

Preconditions

1. The Client Sponsor is authenticated and is the owner of the contract.
2. The contract is `ACTIVE` and the milestone's amount is fully funded in escrow.
3. The Content Creator has submitted a draft for this milestone (milestone state = `SUBMITTED`).
4. The milestone is not in `DISPUTED` state.

Postconditions

On success

1. The milestone state is `PAID` and its sign-off flag is `true`.
2. The escrowed amount is debited from the contract escrow and credited to the Content Creator's platform balance.
3. The Content Creator can download the un-watermarked original asset (per FR-003 the block is lifted).
4. An append-only audit entry records the sponsor identity, amount, gateway reference and UTC timestamp (per NFR-002).
5. Both parties receive a release notification and a downloadable receipt.

On failure

1. The milestone remains `SUBMITTED`; no funds leave escrow; the failure reason is recorded in the audit log.

Main Success Scenario

1. **Client Sponsor** opens the contract and selects the milestone marked `SUBMITTED`.
2. **System** displays the watermarked draft preview, the milestone amount and the version history.
3. **Client Sponsor** reviews the preview and selects **"Approve & Release"**.

4. **System** displays a confirmation dialog stating the amount to be released and that the action is irreversible.
 5. **Client Sponsor** confirms and re-enters their account password (step-up authentication).
 6. **System** verifies that the milestone is `SUBMITTED`, not `DISPUTED`, and that the escrow balance covers the amount.
 7. **System** records the sponsor sign-off and moves the milestone to `APPROVED`.
 8. **System** performs «include» *UC-08 Authorize Escrow Transfer* — it calls the **Payment Gateway** over TLS 1.3 with an idempotency key to move the funds out of the escrow account.
 9. **Payment Gateway** returns an authorization success with a transaction reference.
 10. **System** credits the Content Creator's platform balance and sets the milestone to `PAID`.
 11. **System** lifts the download block on the original asset for the sponsor and writes the audit entry.
 12. **System** notifies both parties and issues a receipt.
 13. Use case ends successfully.
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Alternate Flow — 9a. Payment Gateway Declines the Escrow Transfer

- **9a1.** The Payment Gateway returns a decline or times out after 30 seconds.
 - **9a2.** The System rolls the milestone back from `APPROVED` to `SUBMITTED` in a single transaction, so no partial release is recorded.
 - **9a3.** The System writes a `RELEASE_FAILED` audit entry with the gateway error code (per NFR-002).
 - **9a4.** The System displays "*Payment could not be released — your funds are still safely held in escrow*" and offers **Retry**.
 - **9a5.** The Client Sponsor may retry; the System reuses the same idempotency key so a delayed gateway success is never double-charged.
 - **9a6.** If the transfer fails on 3 consecutive attempts, the System flags the milestone for Platform Admin review, notifies both parties, and the use case ends in failure with all funds still in escrow.
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Alternate Flow — 6a. Milestone Already Disputed («extend» UC-05)

- **6a1.** The System finds the milestone in `DISPUTED` state (a dispute was raised between page load and confirmation).
- **6a2.** The System refuses the release, displays the dispute reason and the raising party.
- **6a3.** Funds remain frozen in escrow and the use case ends without any state change.